

EOD Task Completion Document

MariaDB / MySQL EC2 to Amazon RDS

Date: 17-Sep-2026

Status: Completed

1. Objective

The objective of this assignment was to create MariaDB and MySQL databases on EC2, insert dummy data, take backups, migrate databases to Amazon RDS, configure Multi-AZ, restore a database backup, and create a Read Replica.

2. Tasks Completed

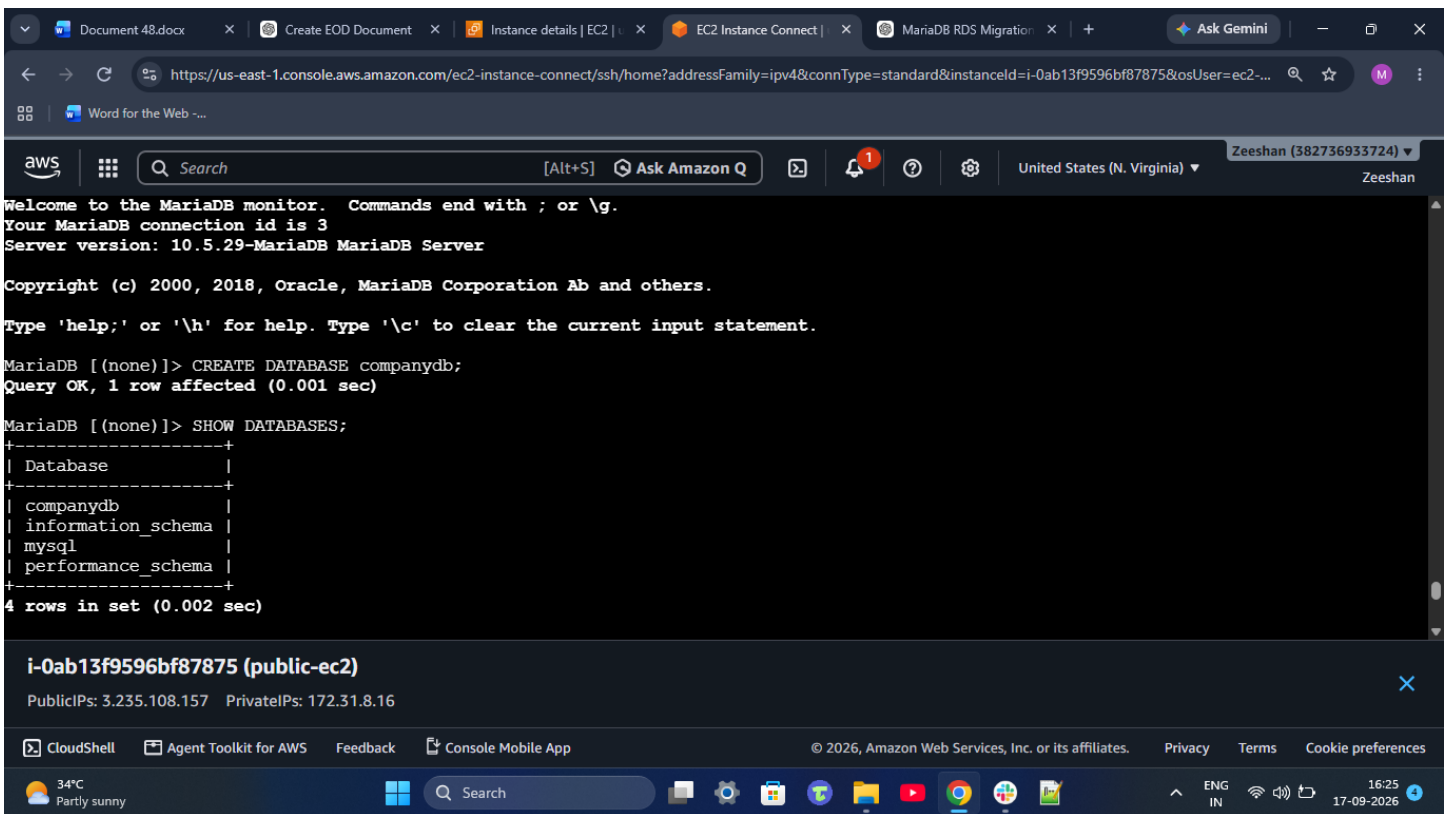
1. Created MariaDB DB on EC2.
2. Inserted dummy data.
3. Took backup of dummy data on EC2.
4. Launched MariaDB RDS instance.
5. Migrated MariaDB database from EC2 to RDS.
6. Installed MySQL DB on EC2.
7. Launched MySQL RDS instance.
8. Configured Multi-AZ.
9. Took backup and restored the database.
10. Created Read Replica.

3. Detailed Steps

Task 1: Create MariaDB DB on EC2

- Launched EC2 instance.
- Installed MariaDB.
- Started MariaDB service.
- Enabled MariaDB service.
- Logged into MariaDB.
- Created database and table.
- Verified the database and table.

Evidence:



Task 2: Insert Dummy Data

- Created sample table.
- Inserted dummy records.
- Used SELECT command to verify the data.

Example:

```
CREATE DATABASE companydb;
```

```
USE companydb;
```

create table

```
CREATE TABLE employees (  
    id INT PRIMARY KEY AUTO_INCREMENT,  
    name VARCHAR(100),  
    department VARCHAR(100),  
    salary INT  
);
```

Insert dummy data

```
INSERT INTO employees (name, department, salary)  
VALUES  
( 'Rahul', 'DevOps', 60000),  
( 'Amit', 'AWS', 70000),  
( 'Zeeshan', 'Cloud', 80000),  
( 'John', 'Testing', 50000);
```

Evidence:

Word for the Web - Mi... Create EOD Document Instance details | EC2 | EC2 Instance Connect | MariaDB RDS Migration | Ask Gemini

https://us-east-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0ab13f9596bf87875&osUser=ec2-...

aws Search [Alt+S] Ask Amazon Q United States (N. Virginia) Zeeshan (382736933724) Zeeshan

Records: 4 Duplicates: 0 Warnings: 0

```
MariaDB [companydb]> INSERT INTO employees (name, department, salary)
-> VALUES
-> ('Rahul', 'DevOps', 60000),
-> ('Amit', 'AWS', 70000),
-> ('Zeeshan', 'Cloud', 80000),
-> SELECT * FROM employees;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MariaDB server version for the right
syntax to use near 'SELECT * FROM employees' at line 6
MariaDB [companydb]> SELECT * FROM employees;
```

id	name	department	salary
1	Rahul	DevOps	60000
2	Amit	AWS	70000
3	Zeeshan	Cloud	80000
4	John	Testing	50000

4 rows in set (0.000 sec)

MariaDB [companydb]>

i-0ab13f9596bf87875 (public-ec2)

PublicIPs: 3.235.108.157 PrivateIPs: 172.31.8.16

CloudShell Agent Toolkit for AWS Feedback Console Mobile App

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Task 3: Take Backup on EC2

Created a database backup using mysqldump.

```
mysqldump -u root -p companydb > companydb_backup.sql
```

Verified the backup:

```
ls -lh companydb_backup.sql
```

Evidence:

A screenshot of the AWS CloudShell interface. The terminal window shows the output of a 'MariaDB dump' command. The output includes the host 'localhost', database 'companydb', and server version '10.5.29-MariaDB'. It lists various system variables like '@@CHARACTER_SET_CLIENT', '@@CHARACTER_SET_RESULTS', '@@COLLATION_CONNECTION', 'utf8mb4', '@@TIME_ZONE', '@@UNIQUE_CHECKS', '@@FOREIGN_KEY_CHECKS', 'SQL_MODE', and 'SQL_NOTES'. Below this, it shows the table structure for 'employees'. At the bottom, a file listing command 'ls -lh companydb_backup.sql' is executed, showing a file 'companydb_backup.sql' of size 2.2K. The AWS console header at the top shows the user 'Zeeshan' and the region 'United States (N. Virginia)'. The bottom status bar shows the temperature '34°C' and the date '17-09-2026'.

Task 4: Launch MariaDB RDS

- Opened **AWS Console** → **RDS** → **Databases**.
- Selected **Create database**.
- Selected **MariaDB** engine.
- Configured:
 - DB instance
 - Storage
 - VPC
 - Subnet
 - Security Group
 - Username/password
- Created the RDS database.
- Waited until the status became **Available**.
- Copied the RDS endpoint.

Evidence:

 MariaDB RDS showing **Available** status.

Task 5: Migrate MariaDB from EC2 to RDS

First allowed EC2 to connect to RDS through the RDS security group on port **3306**.

Tested connection:

```
mysql -h mariadb-rds.c01ec4e8sk2y.us-east-1.rds.amazonaws.com -u admin -p --ssl
```

Created the database on RDS if required.

Imported the backup:

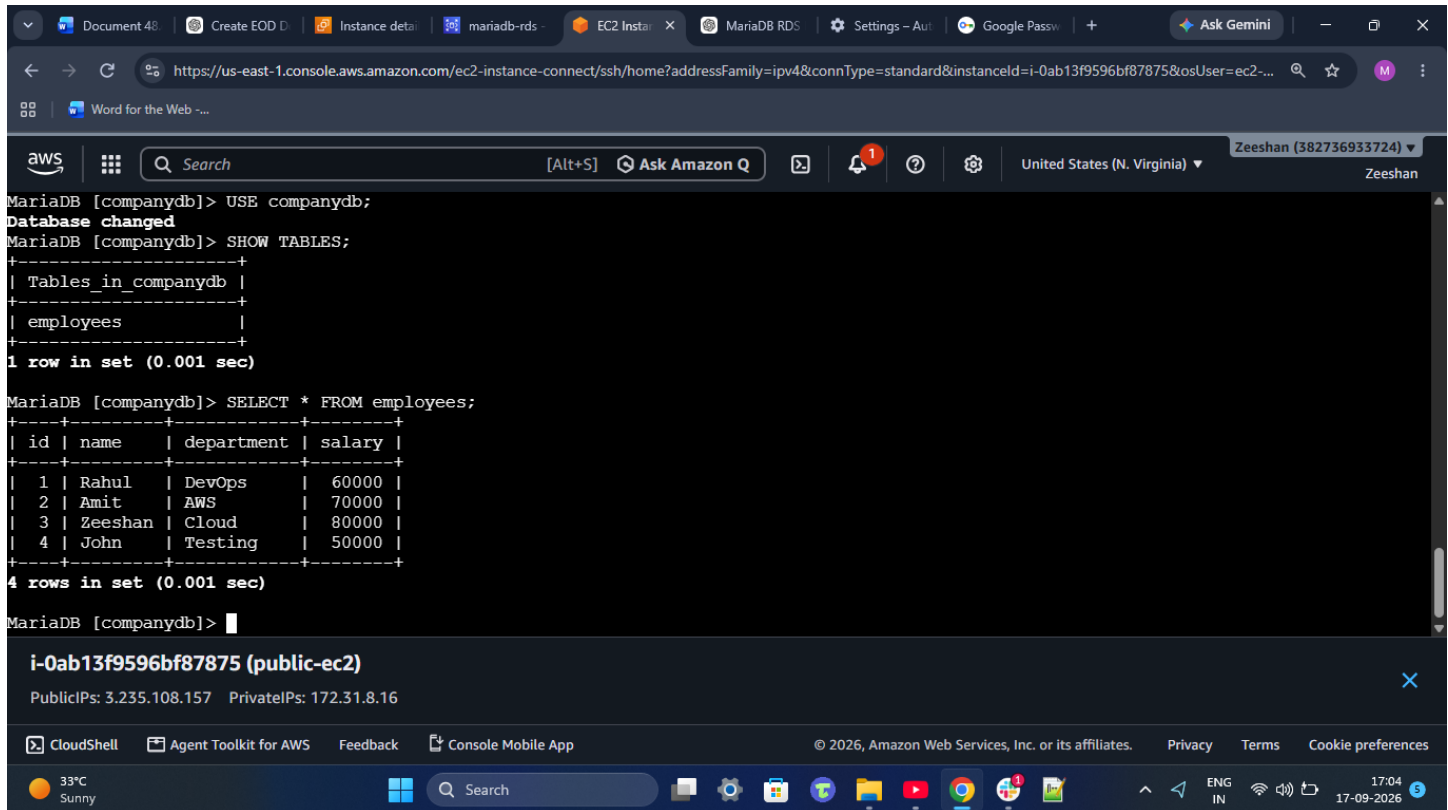
```
mysql -h mariadb-rds.c01ec4e8sk2y.us-east-1.rds.amazonaws.com -u admin -p  
companydb --ssl
```

Verified the migrated data:

```
USE companydb;
```

```
SELECT * FROM employees;
```

Evidence:



The screenshot shows the AWS CloudShell interface with a terminal window. The terminal displays the following commands and output:

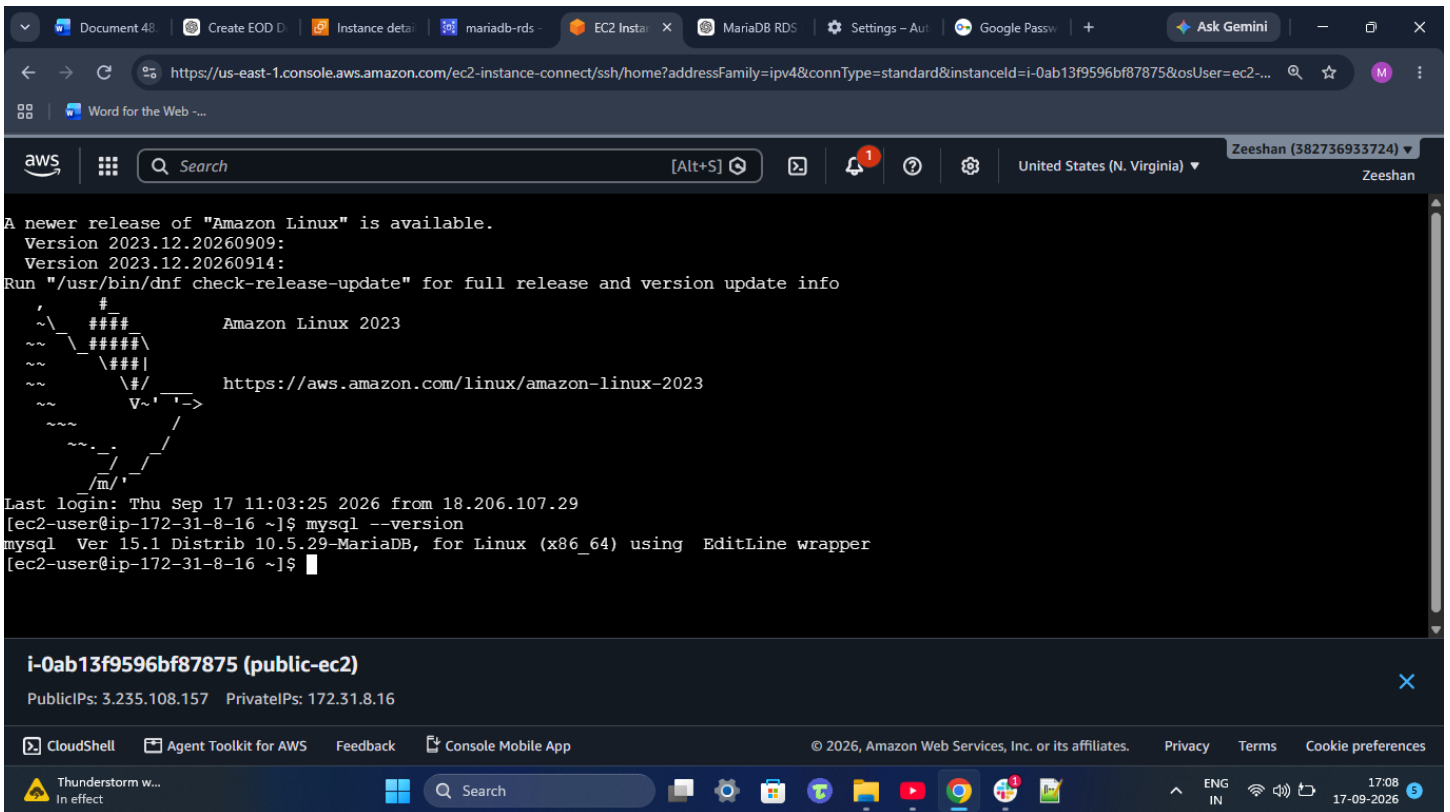
```
MariaDB [companydb]> USE companydb;  
Database changed  
MariaDB [companydb]> SHOW TABLES;  
+-----+  
| Tables in companydb |  
+-----+  
| employees            |  
+-----+  
1 row in set (0.001 sec)  
  
MariaDB [companydb]> SELECT * FROM employees;  
+-----+  
| id | name   | department | salary |  
+-----+  
| 1  | Rahul  | DevOps     | 60000  |  
| 2  | Amit   | AWS        | 70000  |  
| 3  | Zeeshan | Cloud      | 80000  |  
| 4  | John   | Testing    | 50000  |  
+-----+  
4 rows in set (0.001 sec)  
  
MariaDB [companydb]>
```

Below the terminal window, the instance details for **i-0ab13f9596bf87875 (public-ec2)** are shown, including Public IPs (3.235.108.157) and Private IPs (172.31.8.16).

Task 6: Install MySQL DB on EC2

- Installed MySQL on EC2.
- Started MySQL service.
- Enabled MySQL service.
- Verified MySQL version.
- Created sample database/table.
- Inserted test data.

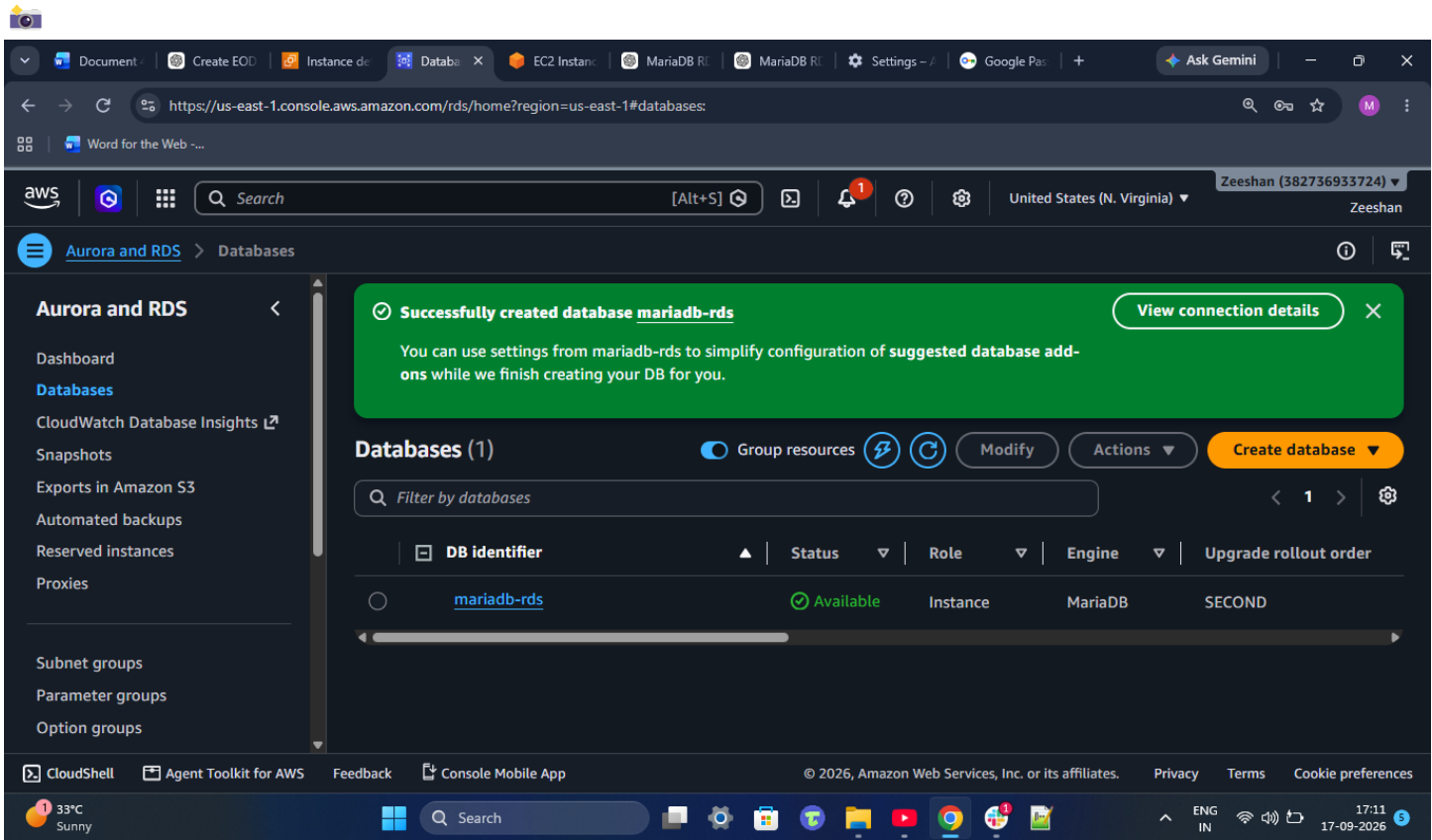
Evidence:



Task 7: Launch MySQL RDS

- Opened **RDS** → **Create database**.
- Selected **MySQL** engine.
- Configured DB instance.
- Configured storage and networking.
- Configured security group.
- Created the RDS instance.
- Waited until status became **Available**.
- Recorded the RDS endpoint.

Evidence:



Task 8: Configure Multi-AZ

- Selected the MySQL RDS instance.
- Clicked **Modify**.
- Enabled **Multi-AZ deployment**.
- Applied the configuration.
- Verified the RDS configuration.

Purpose:

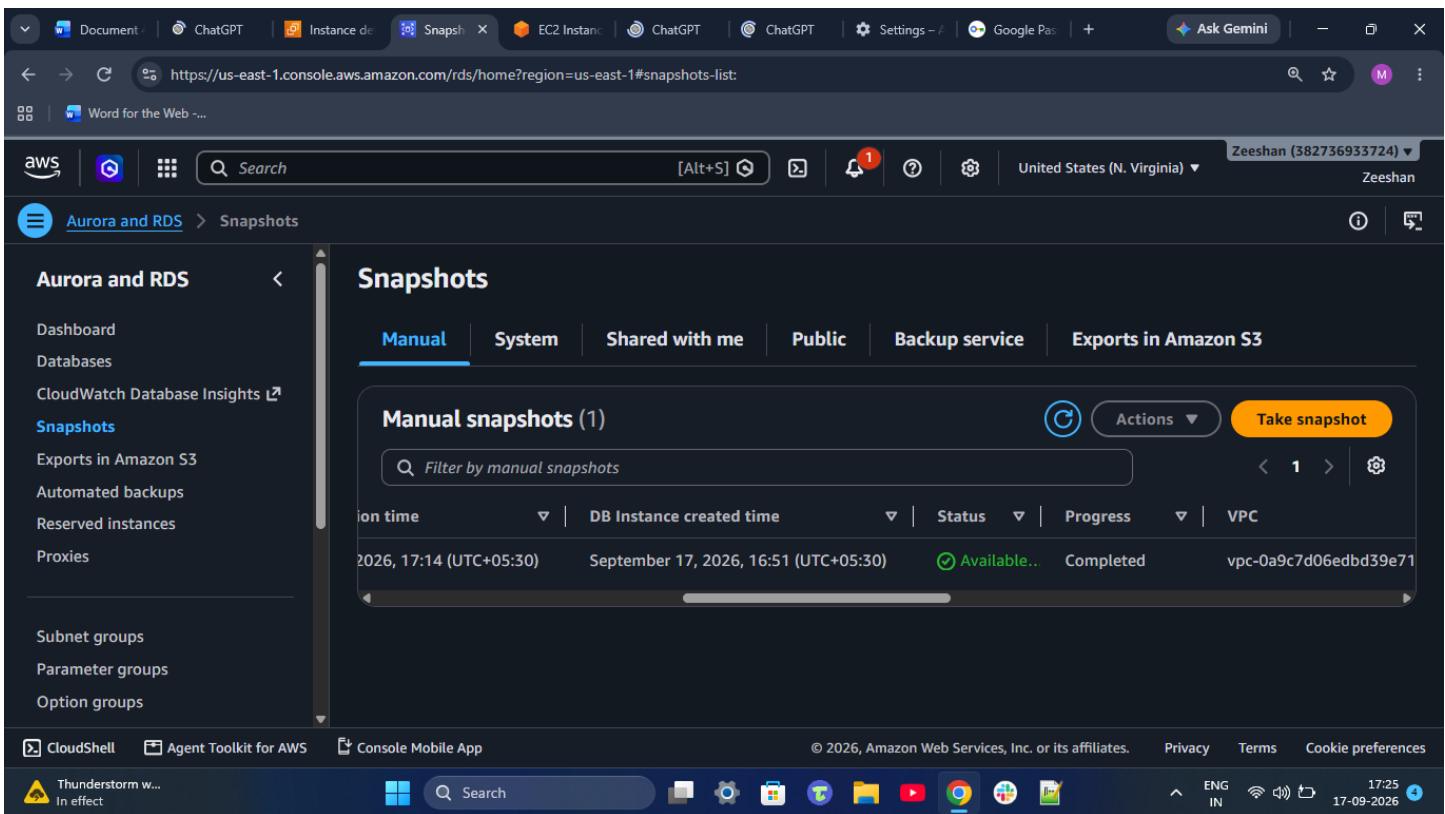
Multi-AZ keeps a standby database in another Availability Zone. It is mainly used for **high availability and automatic failover**.

Evidence:

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Task 9: Backup and Restore Database

- Created an RDS manual DB snapshot.
- Waited for the snapshot to become available.
- Selected the snapshot.
- Selected **Restore snapshot**.
- Created a new RDS database from the snapshot.
- Waited until the restored database became available.
- Connected to the restored database.
- Verified the database and data.



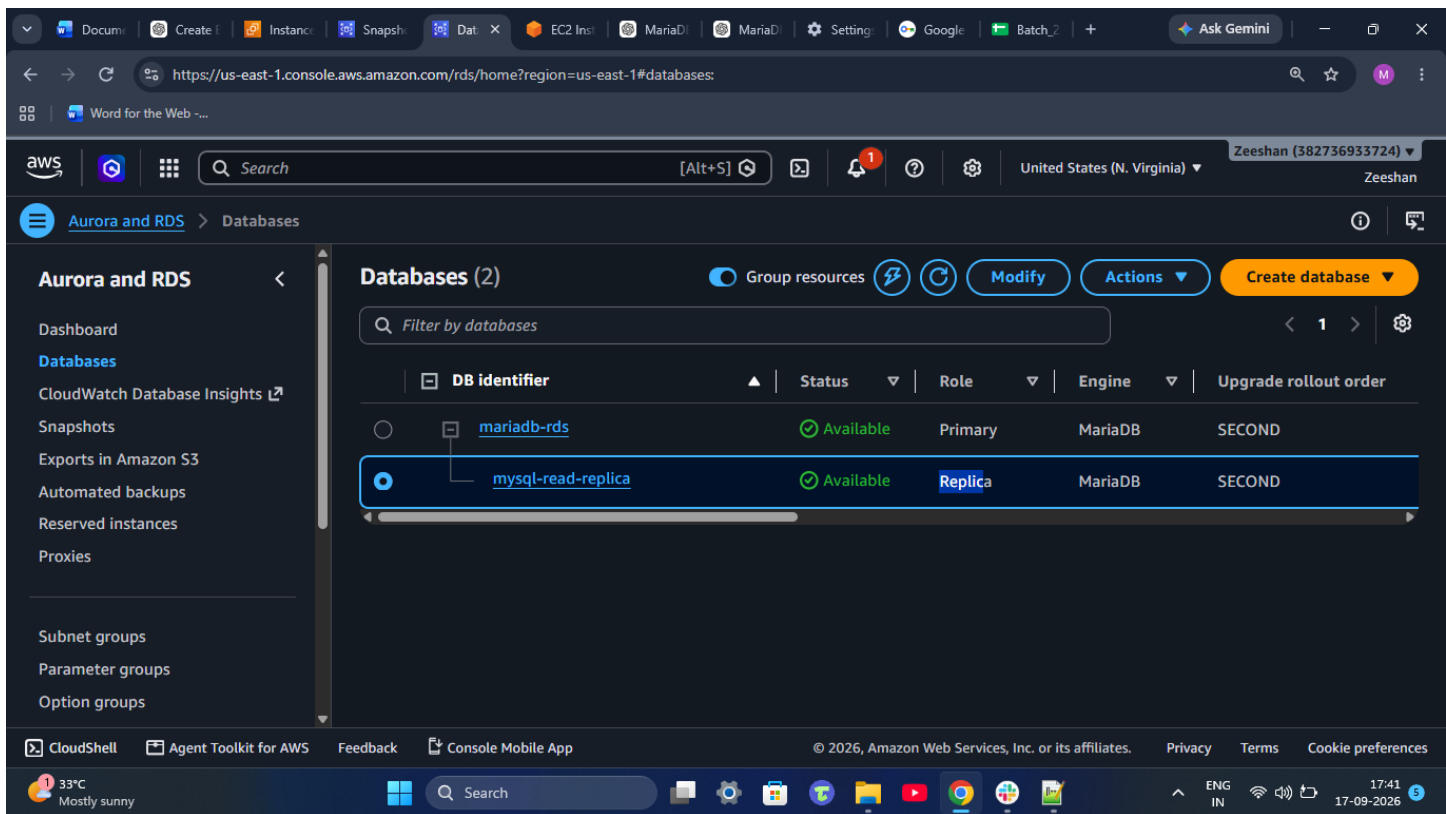
Task 10: Create Read Replica

- Selected the source RDS database.
- Selected **Create read replica**.
- Provided the replica identifier.
- Configured required settings.
- Created the read replica.
- Waited until it became available.
- Checked replication status/monitoring.

Purpose:

A Read Replica maintains a copy of the source database and can be used to handle **read traffic** separately from the primary database.

Evidence:



5. Final Result

The MariaDB database was created on EC2, populated with dummy data, backed up, and migrated to Amazon RDS. MySQL was also installed on EC2 and an Amazon RDS for MySQL instance was created. Multi-AZ was configured, database backup and restore were performed, and a Read Replica was created and verified.